A Consensus-Trained U-Net with Learned Instance Selection on a Threshold Tree

A Solution to the Solar Filament Segmentation Challenge 2026

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ABSTRACT

This report describes our solution to the Solar Filament Segmentation Challenge 2026, a Kaggle competition on automatic segmentation of solar filaments in GONG H-α observations. A single U-Net with a ResNet-34 encoder, run at 1536 px, is trained on the consensus of the annotators and averaged over four flipped views. Its connected components at twelve thresholds form a tree of nested candidates; a gradient-boosted hurdle model, trained on out-of-fold candidates, predicts each one's expected contribution to Panoptic Quality (PQ) from its shape, contrast, position, place in the tree and the disagreement of the four views, and the disjoint set with the largest expected gain is chosen exactly on the tree. A second stage redraws each chosen filament at native resolution in a crop turned along its own axis. On all 707 training images, scored out of fold with a split by calendar month, the pipeline reaches PQ 0.433 (95% bootstrap interval 0.423–0.442), against 0.427 without the second stage, 0.422 with candidates from a single threshold and 0.409 for the same network at 1024 px with a hand-tuned rule; one MAGFiLO annotator scored against another on the same images reaches 0.340. The public leaderboard gives 0.37 for the full pipeline, 0.36 for the first stage alone and 0.35 for the earlier pipelines. We list the variants that did not help and show that most of the remaining loss is disagreement between annotators.

1 INTRODUCTION

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [2]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[1]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [1].

One property of the data shapes every decision: the annotators disagree strongly, and each annotator is scored separately against the same prediction. A prediction can therefore never match all annotators, and the best achievable PQ is far below 1. Our solution is (i) a simple, reproducible segmenter trained on the annotators' consensus; (ii) learned instance selection: candidates from twelve thresholds, a scorer of their expected PQ contribution and an exact choice of the best disjoint set; and (iii) an error analysis, with the annotator-vs-annotator score as a reference, that explains why the usual improvements did not help. Only the competition's training images and annotations are used; no external filament masks, spines or chirality labels. Code, weights and a notebook that reproduces the submission are in the repository, and every experiment is logged in `docs/DESIGN.md`.

[1] www.esairlab.com/

Kaggle Competition, 2026, kaggle.com/competitions/filament-segmentation-2026 2026.

2 METHODOLOGY

2.1 Data and metric

The training set has 707 images (2048×2048 px) with 8,199 annotated filaments; 411 images have one annotator, 145 two and 151 three, giving 1,154 annotator-image entries. The test set has 180 images. Within each entry, a ground-truth (GT) and a predicted mask form a true positive (TP) when their IoU exceeds 0.5, and PQ [7] is $\sum_{TP} \mathrm{IoU}/(|TP| + 0.5|FP| + 0.5|FN|)$, pooled over all entries. It factors into segmentation quality SQ (the mean IoU of the TPs) and recognition quality RQ. Our implementation reproduces the organisers' self-evaluation notebook exactly. That code does not enforce one-to-one matching, so overlapping predictions of one filament would each count as a TP; all our submissions are disjoint by construction.

2.2 Input, target and network

Each image is area-downsampled from 2048 to 1536 px and normalised with fixed constants; there is no other pre-processing. The training target is the consensus, i.e. the fraction of annotators marking each pixel, computed at 2048 px and then downsampled. With binary cross-entropy (BCE) on this soft target the network learns the probability that an annotator marks a pixel, which is the quantity the multi-annotator metric rewards. The network is a U-Net [10] from `segmentation_models_pytorch` [6] with an ImageNet ResNet-34 encoder [5] adapted to one input channel (24.4 M parameters). Loss: BCE + soft Dice. AdamW, learning rate $5 \cdot 10^{-4}$ with one epoch of warm-up and cosine decay, weight decay 10^{-4}, batch 4, 40 epochs. Augmentation: flips, 90-degree rotations, affine (±30°, scale 0.9–1.1), brightness, contrast and gamma. The submitted model is trained on all 707 images (36 min on one NVIDIA A100, about 3 h on an Apple M3 Max).

2.3 Candidates: a tree over thresholds

At inference the network sees four views (identity, horizontal, vertical and both flips); their mean is the probability map and their per-pixel standard deviation (the *view spread*) marks where the model is unsure. Both are upsampled bilinearly to 2048 px. The map is thresholded at twelve levels, 0.25 to 0.80 in steps of 0.05. At each level a morphological closing with a 31 px disk decides which fragments belong to one filament (the masks keep the thresholded pixels), and the 8-connected components of at least 100 px are candidates. Thresholding and closing are monotone, so every component at a higher level lies inside exactly one component at the next lower level: the candidates form a forest. A filament can thus be taken whole at a low threshold, thinner at a high one, or separated at a higher level from a neighbour it merges with lower down (Fig. 1).

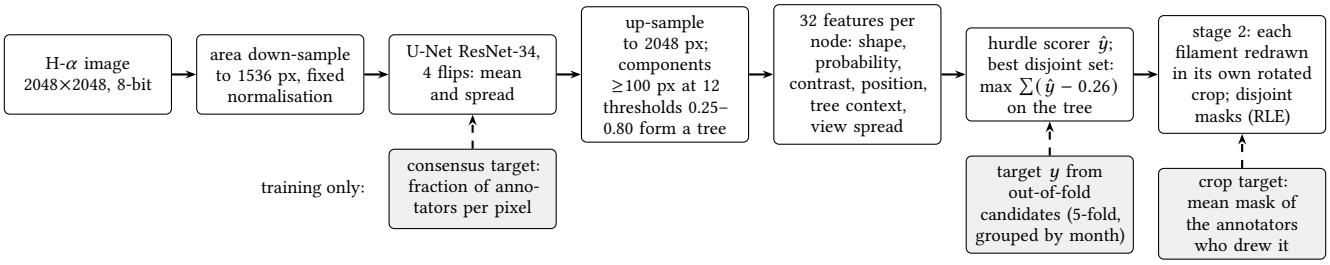


Figure 1: The end-to-end pipeline. White: inference, applied to every test image. Grey: what each learned stage is trained on.

2.4 Instance selection

Every false positive costs half a point in the PQ denominator, so which candidates are submitted matters as much as their shape. A kept candidate that matches k of the n entries of its image adds about $k \cdot \text{IoU}$ to the numerator and $0.5n$ to the denominator either way, so it pays when its expected matched IoU per entry, $y = \frac{1}{n}\sum_e \text{IoU}_e\,[\text{IoU}_e > 0.5]$ over the entries e of its image, exceeds about PQ/2. We predict y for every node of the tree with a hurdle model, $\hat{y} = P(y > 0) \cdot E[y \mid y > 0]$, both parts gradient-boosted trees from scikit-learn [8] (`HistGradientBoosting`, 400 iterations, learning rate 0.05, 31 leaves, at least 40 samples per leaf, L2 1.0), trained on out-of-fold nodes with rows weighted by the image's number of entries, as pooled PQ weights them. The 32 features are: area and log-area; the mean, maximum, standard deviation, 10/50/90% quantiles of the probability inside and the fraction above 0.8; elongation, length, width and compactness; the brightness inside and its contrast with a 15 px ring; distance from the disk centre, latitude and longitude on a disk fitted with a Hough circle; the number of candidates and the mean probability of the image; the node's threshold, its area relative to its parent, its number of children and siblings, the share of its area its children keep, and the number of levels over which it keeps 85% of its area; and the mean, 90% quantile, maximum and relative value of the view spread inside it. The submission is the set of disjoint nodes (an antichain of the forest) maximising $\sum(\hat{y} - \tau)$, which dynamic programming solves exactly: $f(v) = \max(\hat{y}_v - \tau, \sum_c f(c))$ over the children c of v. We use $\tau = 0.26$, the middle of a range (0.24–0.28) over which OOF PQ varies by 0.0004.

2.5 Stage 2: refinement in rotated crops

A level set of one map cannot draw every filament: a selected node may be a fragment, or thinner or wider than the annotators drew it. Following the second stage of a public notebook [11], each selected filament (the *seed*) is redrawn in a 384×128 px crop centred on it, its principal axis turned horizontal, covering 1.5 times its length at native resolution (scaled down only when it does not fit). The four input channels are the image, the stage-1 probability, the seed and the other seeds of the image. The target is the mean of the masks drawn for this filament by the annotators who drew it (each entry's mask that overlaps the seed most, if its IoU exceeds 0.3); whether the filament exists stays stage 1's decision. A U-Net with a ResNet-34 encoder is trained per fold for 30 epochs on the other folds' crops (BCE + soft Dice, flips; 16–20 min per fold on an M3 Max); at test time the five fold models and four flips are averaged. The output is thresholded at 0.40, fixed before the evaluation, and pasted back: only pieces touching the seed are kept, a seed covered at least half by its neighbours' new masks is absorbed, and overlaps go to the higher refined probability, so the masks stay disjoint. Inference with both stages takes 6.6 s per image on an M3 Max.

2.6 Validation split and tuning

Filaments persist for days, so images from the same week are near-duplicates. We use a 5-fold `GroupKFold` grouped by calendar month (YYYYMM), so no month is in both training and validation. The threshold and closing are grid-searched with the exact metric on the pooled out-of-fold (OOF) probabilities of the five fold models. The scorer is cross-validated on the same folds: the scorer applied to a fold's nodes is trained on the other four, so every OOF number below comes from models that never saw the image. The view spread of the OOF images is computed by each fold's model, as at test time. All choices were made on OOF, never on the public leaderboard, which covers about 90 test images and shows two decimals.

3 EVALUATION

3.1 Main results

Table 1 gives the main numbers. Uncertainty comes from two sources: the spread of PQ over the five folds, and a bootstrap that resamples whole images (all entries of an image together, 2,000 resamples) and recomputes the pooled PQ. The paired bootstrap of the difference to the previous pipeline resamples the same images for both. To put the numbers in scale, each annotator's masks are scored as predictions against every other annotator of the same image (296 images with two or three annotators).

The gains accumulate in steps that each hold up in a paired bootstrap. Resolution and views: tuned per fold, 1536 px beats 1024 px on four of five folds (+0.009, −0.002, +0.009, +0.005, +0.007), and 2048 px matches 1536 px on the two folds tried. Scoring: with a single threshold the scorer beats the submitted hand rule (confidence ≥ 0.8) on every fold, by 0.002–0.009, but the best pooled hand rule (confidence ≥ 0.85, PQ 0.420) by only 0.002. Tree: candidates from six thresholds with the hurdle model and the context features give +0.004 [0.001, 0.008]; twelve thresholds alone add nothing (0.425), and the view-spread features bring the gain to +0.005 [0.001, 0.008] over the single threshold, on four of five folds (per fold 0.434, 0.425, 0.411, 0.430, 0.434 against 0.432, 0.415, 0.407, 0.421, 0.435). Against the 1024 px rule stage 1 gains +0.018 [0.013, 0.023], on every fold.

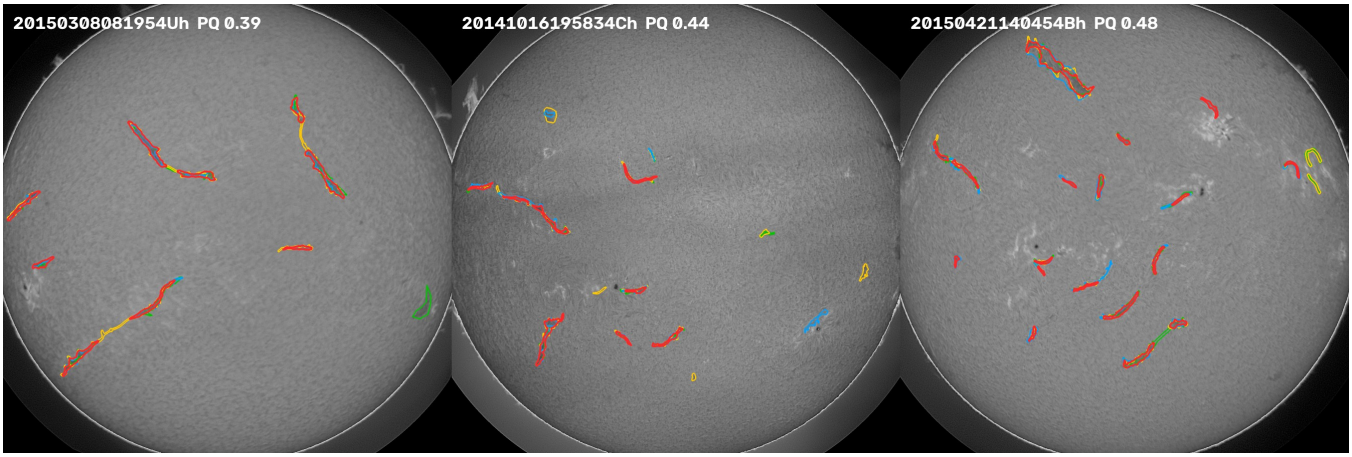


Figure 2: Out-of-fold predictions of stage 1 (red) on three validation images with three annotators (green, yellow, blue), at the 50th, 75th and 90th percentile of image PQ among images with at least five filaments per annotator. Many misses are filaments marked by one annotator only.

Table 1: Main results. OOF: out of fold, 707 images, 1,154 entries. Brackets: 95% bootstrap interval over images.

	PQ	SQ	RQ
Final (stage 2), OOF	**0.433** [0.423, 0.442]	0.677	0.639
per fold, mean ± s.d.	0.433 ± 0.009		
296 multi-annotator images	0.434	0.674	0.644
Annotator vs. annotator (same 296)	0.340	0.634	0.536
1024 px, 2 views, hand rule	0.409 [0.399, 0.418]	0.666	0.614
+ 4 views	0.412	0.667	0.617
+ component scorer	0.417	0.668	0.624
1536 px, 4 views, hand rule	0.417	0.671	0.621
+ scorer, one threshold	0.422 [0.412, 0.431]	0.674	0.626
+ tree (6 levels), hurdle	0.426	0.676	0.630
+ 12 levels, spread (stage 1)	0.427 [0.417, 0.436]	0.675	0.632
+ stage 2 (final)	0.433	0.677	0.639
Stage 2 over stage 1 (paired)	+0.006 [0.003, 0.009]		
Stage 1 over one threshold (paired)	+0.005 [0.001, 0.008]		
Stage 1 over 1024 px rule (paired)	+0.018 [0.013, 0.023]		
Public LB, stage 2 (final)	0.37	–	–
Public LB, stage 1	0.36	–	–
Public LB, earlier pipelines	0.35	–	–

Stage 2: each fold's crops are redrawn by the refiner that did not see them, and PQ rises on all five folds (0.434, 0.433, 0.419, 0.442, 0.437), by +0.006 [0.003, 0.009] pooled, through RQ (0.632 to 0.639) more than SQ (0.675 to 0.677): redrawn fragments now cross IoU 0.5. The pixel threshold barely matters (0.433 at 0.40 and 0.45, 0.432 at 0.35 and 0.50). The leaderboard moves by one step at its two decimals with each stage (0.35 to 0.36 to 0.37), consistent with but not evidence for the gain: scaled to the 90 or so public images, the bootstrap interval of a single pipeline is about ±0.027, larger than any of these gains. Local OOF ran 0.06–0.07 above the public leaderboard for every submission, a gap other entrants report too.

Fig. 3 shows the three evaluation plots requested by the organisers on the OOF predictions of stage 1. Over GT/prediction pairs with IoU > 0, the mean IoU is 0.575 and the mean Dice 0.708 (0.553 and 0.688 for the 1024 px rule pipeline). 3.1% of predicted filaments overlap more than one annotation (merges) and 3.8% of annotations overlap more than one prediction (splits); 10.6% of predictions overlap no annotation of their entry and 22.7% of annotations overlap no prediction. Compared with the hand rule, learned selection removes about a quarter of the false positives; the tree then recovers some of the annotations a single threshold misses or merges. Fig. 2 shows typical images.

3.2 What did not help

Table 2 lists the variants we tried, each with its own tuned post-processing. A Mask R-CNN [4] detector learns grouping explicitly, but its masks are coarser (SQ 0.642 vs. 0.657 for a U-Net trained for a similar number of epochs); adding its extra instances, or an axis-aligned crop refiner around each instance following Reche & Cid [9], changes PQ by less than 0.002 on the fully trained U-Net. A ResNet-50 encoder, an input divided by its own radial brightness profile (limb-darkening correction) and a Lovász hinge loss [3] match or underperform the baseline; with the corrected input the best threshold moves from 0.45 to 0.65, so the network becomes more confident but not more accurate. Averaging two equal-strength models with different encoders gives the PQ of the better one, and the 5-fold average gives the same leaderboard score as one model: the models share their errors because the errors come from the annotators, so stage 1 is one network.

3.3 Where the remaining PQ is lost

Near misses. This analysis uses the 1024 px rule pipeline. Of its 3,174 OOF false negatives, 45% overlap no prediction, 18% overlap one at IoU < 0.3 and 37% at IoU 0.3–0.5. Among the near misses the prediction is smaller than the GT in 46% of cases and larger in 34%; only 17% are merges with another filament and 16% splits.

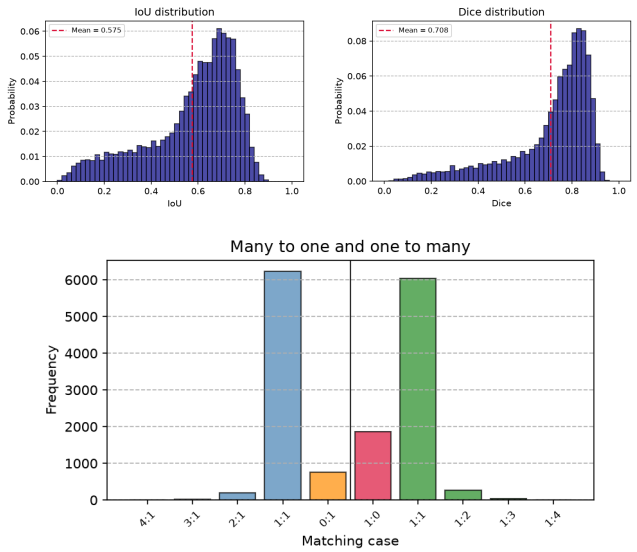


Figure 3: Out-of-fold IoU and Dice distributions over GT/prediction pairs with IoU $>$ 0, and the one-to-many / many-to-one counts (left: number of GT linked to each prediction; right: number of predictions linked to each GT).

Table 2: Variants (fold 0, 142 images, tuned PQ) and post-processing rules (5-fold OOF, 707 images).

Variant (fold 0)	PQ
U-Net ResNet-34, 40 epochs (baseline)	**0.416**
10 / 20 epochs	0.389 / 0.396
+ axis-aligned crop refiner	0.413
+ Mask R-CNN instances not covered	0.417
Mask R-CNN alone (disjoint masks)	0.377
ResNet-50 encoder	0.414
Limb-darkening-corrected input	0.411
BCE + Lovász hinge	0.390
Average of ResNet-34 and ResNet-50	0.416
8 dihedral views (4 flips: 0.419)	0.419
2048 px, folds 0 / 3 (1536 px: 0.403 / 0.387)	0.404 / 0.388
Post-processing rule (OOF, 1024 px)	**PQ**
Submission setting (confidence $\geq$ 0.8)	0.409
Confidence $\geq$ 0.85	0.411
No closing, no confidence or limb filter	0.398
Size-dependent confidence threshold	0.411
Hysteresis (seed and grow thresholds)	0.409
Masks dilated / eroded by 1 px	0.381 / 0.399
70-degree limb cut-off	no change

Dilating or eroding every mask by one pixel costs 0.01–0.03 PQ, because filaments are only a few pixels wide: the extent errors are per instance, not a bias a rule could correct. *Small filaments.* Recall falls to 0.36 below 500 px. Annotators rarely agree on them: the share of other annotators marking the same filament (IoU $>$ 0.5) is 0.28 below 300 px and about 0.60 above 1,000 px. *When a doubtful prediction pays.* A prediction matching k of n annotators raises PQ only if $k/n > \mathrm{PQ}/(2\,\mathrm{SQ}) \approx 0.31$. A filament marked by one annotator of three is break-even, which is why lowering the area or confidence thresholds never helped; 28% of our false positives match another annotator's filament on the same image.

3.4 Strengths, weaknesses and possible improvements

Strengths. One network, two small tree models and five crop refiners, no external data, 6.6 s per image, and a submission reproduced byte for byte by the notebook. Out of fold it exceeds a single annotator on both boundary quality (SQ) and detection (RQ): trained on the consensus, it predicts the filaments most annotators agree on. Selection optimises the metric directly: the scorer estimates each candidate's expected contribution, the tree lets it choose the threshold per filament, and the exact antichain keeps the masks disjoint without hand-tuned rules. *Weaknesses.* Missed annotations are the largest error (23% of annotations overlap no prediction), mostly small filaments and filaments marked by one annotator; extent errors of one or two pixels on thin filaments cost SQ; and every gain since the 1024 px rule is below the resolution of the public leaderboard, so the choice rests on the paired out-of-fold comparisons. *Improvements.* Most of the remaining loss is disagreement between annotators, which no segmenter removes. Within reach are more diverse members for the probability map (other encoders and decoders, whose disagreement would also sharpen the spread features) and for the refiner.

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